

enterprises. Margarita Island would probably have been of interest to such groups only as a vacation destination. The more common adult-onset diseases, such as cardiovascular disease, psychiatric disorders and cancer, provide ready targets for pharmaceutical development or gene therapy, and so it becomes less and less likely that

'big pharma' will take an interest in investigating the causes of birth defects. □

1. Fogh-Andersen, P. *Inheritance of Harelip and Cleft Palate* (Nyt Nordisk Forlag, Copenhagen, 1942).
2. Braybrook, C. et al. *Nature Genet.* **29**, 179–183 (2001).
3. Moore, G.E. et al. *Nature* **326**, 91–92 (1987).
4. Sözen, M.A. et al. *Nature Genet.* **29**, 141–142 (2001).
5. Suzuki, K. et al. *Nature Genet.* **25**, 427–430 (2000).
6. Bamshad, M. et al. *Nature Genet.* **16**, 311–315 (1997).
7. Lidral, A.C. *Am. J. Hum. Genet.* **63**, 557–568 (1998).
8. Li, Q.Y. et al. *Nature Genet.* **15**, 21–29 (1997).
9. Schinke, M. & Izumo, S. *Nature Genet.* **27**, 238–240 (2001).
10. Tada, M. & Smith, J.C. *Dev. Growth Differ.* **43**, 1–11 (2000).
11. Beaty, T.H. *Ann. Epidemiol.* **11**, 434–442 (2001).
12. van den Boogaard, M.J.H., Dorland, M., Beemer, F.A. & van Amstel, H.K.P. *Nature Genet.* **24**, 342–343 (2000).
13. Schutte, B. C., et al. *Genome Res.* **10**, 81–94 (2000).

Islands of linkage disequilibrium

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A detailed knowledge of patterns of linkage disequilibrium in human populations is widely seen as a prerequisite for effective population-based disease gene mapping. New data suggest that linkage disequilibrium is highly structured into discrete blocks of sequence separated by hot spots of recombination.

The natural world is not famous for making life easy for human geneticists. Despite application of an increasingly powerful set of tools provided by the Human Genome Project, the complexity of common diseases has made them largely refractory to genetic analysis. In the face of this complexity, geneticists agree that the family-based approaches that proved so successful for the monogenic diseases are not up to the job. Instead, most favor association studies, in which genetic and phenotypic variation is compared in large population samples in order to identify correlations implicating genetic risk factors. The classic example is the case-control study in which, for example, a group of sufferers from a condition is compared to a group of healthy controls.

Linkage disequilibrium in gene mapping

Association studies, however, introduce a new complication. Linkage disequilibrium (LD) is the non-random association between alleles at different loci, and it creates opportunities as well as difficulties in gene mapping. In the extreme case, an allele found at one locus predicts which allele would be found at the other, making one of the loci redundant for mapping purposes. Thus, it should, in principle, be possible to use knowledge of LD to design strategies that could represent most of the variation in the genome by genotyping only a small fraction of the polymorphic

sites. On the other hand, it is difficult, and sometimes impossible, to pick out the causal variants in a set of sites in strong LD using only association data (see figure).

For these reasons there has been an explosion of interest in patterns of LD in human populations. In an ideal world the patterns would be highly consistent, in which case it would be relatively straightforward to figure out how to make LD work for us and not against us. Early signs, however, were not encouraging. For example, Clark *et al.*¹ suggested that extremely variable patterns of LD over relatively short sequence stretches, such as they observed in the lipoprotein lipase gene (*LPL*), might seriously complicate association studies. A practical example of the difficulty of separating causation and correlation comes from an association study that implicated a promoter variant in the gene *CYP3A5* (which encodes a member of the cytochrome P450 IIIA subfamily) as determining polymorphic activity of liver *CYP3A5* (ref. 2). A subsequent sequencing study of the *CYP3A* locus, however, demonstrated that the implicated variant was in the promoter region of a *CYP3A* pseudogene (*psI*) about 25 kb upstream from *CYP3A5*, and so was very unlikely to be the cause of the polymorphic activity³.

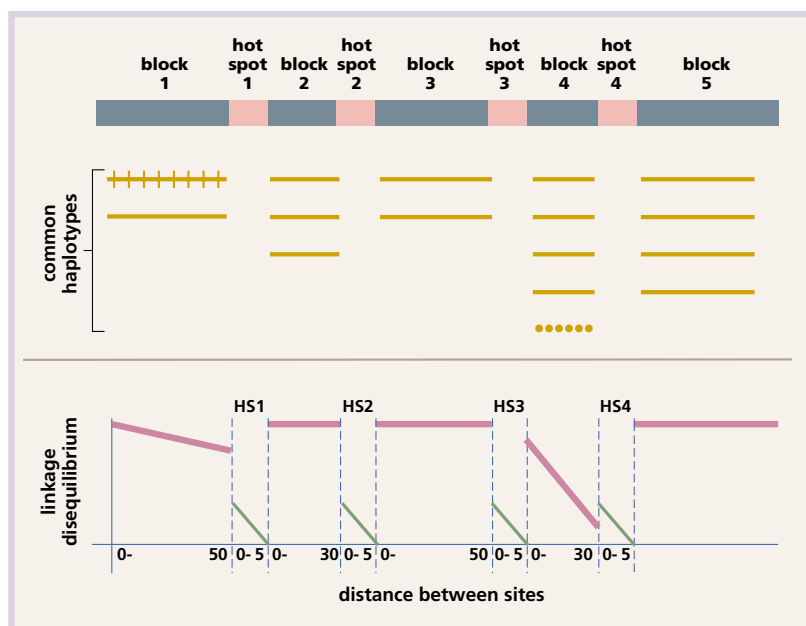
In coarser assessments of broad patterns of LD, both in genes and in non-coding regions, the overall picture has been one of inconsistency, with patterns differing

markedly across genomic regions and across populations and failing to show a consistent dependence on genetic distance. For example, Reich *et al.*⁴ studied 160 kb at each of 19 different regions, documenting consistently maximal LD across some of the regions, highly variable patterns across others, and large average differences between African and north European populations.

In fact, such complexity is not unexpected. Factors such as selection and local mutation and recombination rates can influence LD in particular genomic regions, and population factors such as changes in population size and admixture between differentiated populations can influence genome-wide patterns of LD. Although extreme variability of LD would not preclude mapping genes with association data, it would certainly complicate the job of obtaining a detailed description of genome-wide patterns of LD.

Recombination hot spots

Two lines of evidence published in this issue now suggest that human LD might not be so complicated after all. Characterizing a 500 kb stretch of chromosome 5q31, Mark Daly and colleagues⁵ (page 229) note that the pattern of LD is highly structured, with stretches of consistently high LD interspersed with short intervals of rapid breakdown. In fact, the 500-kb interval can be divided into 11 blocks of consistently high LD. In spite of including more than 75% of



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The lowdown on LD. Idealized representation of block-like structure of linkage disequilibrium, with regions of low haplotype diversity separated by recombinational hot spots. Lines below the blocks represent examples of the number of common haplotypes that might be present for such blocks. SNPs distinguishing the two common haplotypes in block 1 are represented by short vertical lines. The graphs plot (idealized) LD as a function of distance, averaged across pairs of sites, either for sites within a given block or within a hot spot. The plots show that within a block LD decays only gradually with distance, or not at all. Within hot-spot areas, however, LD falls away much more rapidly with distance. If no LD-generating event, such as a bottleneck, has recently occurred in the population, then there may have been enough recombination across the hot spots that the haplotypes in adjacent blocks are randomly associated. Similarly, with sufficient time, or in blocks with higher within-block recombination rates, LD may be substantially reduced for distant sites within a block, as represented here in block 4. Note that for block 1, any of the SNPs indicated would be sufficient to represent the majority of the haplotypic variation within this block. If haplotype 1 were shown to increase the risk of a condition relative to haplotype 2, however, it would be impossible to determine from association data which of the SNPs distinguishing haplotypes 1 and 2 was the biological cause of the increased risk.

the total sequence, the blocks show no evidence for recombination and extremely little haplotype diversity.

Daly *et al.*⁵ suggest that the blocks of LD result from the localization of recombination to irregularly spaced hot spots, an explanation given dramatic support by the study of Alec Jeffreys and colleagues⁶ (page 217) on LD and recombination in the HLA region. A high-resolution analysis of LD throughout 216 kb of the HLA class II region again shows clear block-like structure. Estimation of recombination frequencies in sperm reveals three clusters of recombinational hot spots, apparently accounting for at least 94% of the observed recombinations and corresponding to areas of LD breakdown. Moreover, within these areas, the fine-scale pattern of LD corresponds to the precise boundaries of hot spots 1–2 kb long.

As noted by Daly *et al.*⁵, this atomistic picture of LD is largely consistent with published data. For example, the plot of LD against distance averaged across the 19 regions studied by Reich *et al.*⁴ shows a gradual decay with distance, but this gradual decay is not recapitulated within regions. In fact, on visual inspection, at least half of the regions show precisely the pattern noted by Daly *et al.*⁵, with LD appearing to collapse at a specific distance, whereas most of the others give no clear impression. In the case of *LPL*, Templeton *et al.*⁷ had already noted that recombination is apparently mainly restricted to a 1.9-kb segment and that haplotype diversity is again very simple outside this segment.

Even in this atomistic world there does remain a quantitative aspect to patterns of LD, both in inter-block disequilibrium, observed in the 5q region by Daly *et al.*⁵, and in the possibility of gradual decay of LD within blocks, observed in the HLA region by Jeffreys *et al.*⁶ but not in 5q, possibly as a result of the greater genealogical depth for the HLA region or greater within-block recombination rates.

The greatly reduced LD in Africa reported by Reich *et al.*⁴ and others could therefore result from less inter-block LD, sharper decay of disequilibrium within blocks or some combination of these two. Indeed, at the extreme, if a population has been stable and of large size for sufficiently long, then there could be enough recombination within some blocks to reduce LD between distant markers within blocks to unmeasurable levels. In this case, the clear block structure reported in Europeans could be absent from other populations and/or from certain regions in which within-block recombination rates are sufficiently high (for example, 'block' 4 in the figure).

Implications for association studies

A largely atomistic pattern of LD would have profound implications both for the description of LD and for typing strategies and data analysis in association studies. With respect to LD, the goal would largely switch from the assessment of its quantitative dependence on genetic distance to the detection of block boundaries and the coupling strengths of

associations across them. With blocks defined, a relatively small number of markers could represent most of the haplotype diversity in the genome. Covering hot-spot regions, however, would considerably increase the number of markers required relative to the amount of sequence represented by hot spots⁶. For example, Daly *et al.*⁵ note that in their first block of LD, spanning 84 kb, 96% of the chromosomes sampled have one of two haplotypes. Just one single-nucleotide polymorphism (SNP) would therefore be sufficient to represent most of the diversity in this 84-kb stretch in Europeans, while in other blocks a handful at most would be required (because of the ascertainment process, everything discussed here relates to common SNPs). The observed pattern of LD also means, however, that association data will generally be insufficient to pinpoint the precise causal SNP within blocks of high LD.

These implications of the block structure of LD are clearly illustrated in the study of John Rioux and colleagues⁸ (see page 223) on a risk factor for Crohn disease in the 5q31 region. The simple haplotype structure throughout the region provides clear and strong evidence that one of the haplotypes is a risk factor for Crohn disease, with an increased risk of 2.0 in heterozygous form. The at-risk haplotype extends over 250 kb, however, and the association data provide no means of selecting the SNP that is responsible for increased risk out of the many SNPs that are uniquely associated with the risk-conferring haplotype.

It is worth noting that the existence of discrete blocks of consistent LD would probably reduce the need for sophisticated population-genetic inference in gene mapping. Instead of using formal models to estimate statistically the precise location of a causal variant, the focus would shift to simple assignment of detected effects to a hot spot—delimited block. Fine assignment within the block would rely on the biology of the trait or, where possible, on cross-population comparisons looking for either rare discrete recombinants or ‘old’ populations in which LD has had time to decay even within blocks. For example, the CYP3A mystery was resolved when an intronic splice-site polymorphism in *CYP3A5* was shown to be in complete association with the *ps1* SNP. The *CYP3A5* variant was the clear choice on biological grounds, but the inference was supported by analysis of enzymatic activity in another ethnic group in which the association between the two SNPs was incomplete⁹. This underscores the importance of describing haplotype structure in multiple populations.

Despite its limited scope, the current evidence is compelling and implies that studies should now assume that LD has a block-like structure until such a view either is con-

firmed or becomes unsustainable. But that still leaves difficult questions about priorities and strategies in characterizing global patterns of LD. For example, one discouraging note in the current batch of papers comes from the study of John Todd and colleagues¹⁰ (page 233), in which 135 kb spanning 9 genes are characterized. They observed long stretches of LD with very limited haplotype diversity, but found that the SNPs currently available in dbSNP (<http://www.ncbi.nlm.nih.gov/SNP>) are not generally sufficient to capture that diversity. Todd and colleagues¹⁰ propose a systematic program to search for SNPs in exons and in small upstream and downstream regions around genes (compare ref. 11). Daly *et al.*⁵, on the other hand, clearly envisage a genome-wide effort.

I can see some argument for doing gene regions first. But I think it is clear that the whole genome must be characterized, given our ignorance of, for example, the distance between enhancer elements and the genes they influence. Ignoring currently available SNPs when determining the haplotype structure of genes seems wasteful, however. An inspection of dbSNP makes clear the tremendous variability of currently available SNPs per kilobase in different genes.

It seems likely, therefore, that available SNPs would be sufficient to capture haplotype diversity for some blocks but not others, and that a more efficient strategy would be to characterize available SNPs in targeted regions first, followed by SNP discovery as necessary where the available SNPs appear insufficient to capture the haplotype diversity.

Association studies suddenly look much less difficult than before, and the case for an international project explicitly dedicated to describing haplotype structure in multiple populations now seems overwhelming. It is essential that such a project be as inclusive as possible to ensure that the results can be used to study conditions prevalent in any and all human populations. □

1. Clark, A.G. *et al.* *Am. J. Hum. Genet.* **63**, 595–612 (1998).
2. Paulussen, A. *et al.* *Pharmacogenetics* **10**, 415–424 (2000).
3. Finta, C. & Zaphiropoulos, P.G. *Gene* **260**, 13–23 (2000).
4. Reich, D.E. *et al.* *Nature* **411**, 199–204 (2001).
5. Daly, M.J. *et al.* *Nature Genet.* **29**, 229–232 (2001).
6. Jeffreys, A.J. *et al.* *Nature Genet.* **29**, 217–222 (2001).
7. Templeton, A.R. *et al.* *Am. J. Hum. Genet.* **66**, 69–83 (2000).
8. Rioux, J.D. *et al.* *Nature Genet.* **29**, 223–228 (2001).
9. Kuehl, P. *et al.* *Nature Genet.* **27**, 383–391 (2001).
10. Johnson, G.C.L. *et al.* *Nature Genet.* **29**, 233–237 (2001).
11. Stephens, J.C. *et al.* *Science* **20**, 489–493 (2001).